TAM-FNO: Temporally Affine-Modulated Fourier Neural Operator for Dynamic Underwater Acoustic Field Prediction

Yufei Yang, Yifan Sun, Lei Cheng, and Peter Gerstoft

***Abstract*—Fourier Neural Operators (FNO) have emerged as a powerful data-driven alternative to conventional numerical solvers for rapid underwater acoustic field prediction. Recent work in this area primarily focuses on optimizing spatial network architectures for static environments. However, most existing models emphasize spatial mappings and pay limited attention to temporal thermodynamic variations in dynamic ocean acoustic fields. In this work, we propose the Temporally Affine-Modulated Fourier Neural Operator (TAM-FNO) to address this limitation. Instead of relying on straightforward spatio-temporal concatenation, we introduce a dual-track architecture that leverages Feature-wise Linear Modulation (FiLM) for spatio-temporal decoupling. This design allows temporal information to modulate intermediate spatial feature channels without changing the spatial input format. Numerical experiments on a South China Sea dataset show that TAM-FNO substantially reduces prediction error compared with baseline methods and remains robust across seasonal transitions.**

***Index Terms*—Fourier Neural Operator, Underwater Acoustics, Spatio-Temporal Decoupling, Feature-wise Linear Modulation**

I. Introduction

UNDERWATER acoustic field prediction is essential for downstream tasks such as environment-aware sensor placement optimization and autonomous vehicle path planning [1], [2]. Conventional methods perform these predictions by repeatedly running computationally expensive numerical solvers, such as Range-dependent Acoustic Model, RAM [3], under varying environmental configurations. These numerical approaches, while accurate, often struggle with real-time requirements [4]–[6]. To overcome this computational bottleneck, data-driven approaches, fueled by broader successes of machine learning in acoustics [7], [8], have emerged. Particularly, Fourier Neural Operators (FNO) [9], [10] and their extensions [11]–[14] have become powerful surrogate models for rapidly predicting transmission loss (TL) from sound speed field (SSF). Recent variants have further integrated physics-informed constraints [15], [16] to improve generalization in acoustic charting [17]–[19], and applied them to broader wave propagation and seismic inversion problems [20]–[27].

Despite their efficiency, existing FNO applications in underwater acoustics largely assume a static ocean environment. In reality, the ocean is highly dynamic, heavily influenced by long-term thermodynamic variations such as seasonal temperature and salinity changes. Therefore, introducing temporal awareness is essential. The most intuitive approach to achieve this is to concatenate the time variable t (as a scalar or encoding) directly onto the input SSF along the channel dimension [9], [28]. However, direct temporal concatenation does not reliably exploit temporal information. In our experiments, scalar concatenation provides only marginal change relative to the static baseline, while Fourier-feature concatenation may degrade performance. This may occur because broadcasting a global temporal descriptor across a spatial grid introduces spatially uniform signals with dominant low-frequency content, which can disturb the subtle spatial variations crucial for accurate wavefield prediction.

In this work, we propose TAM-FNO, an architecture for acoustic field prediction in dynamic ocean environments. Since predicting acoustic propagation under time-varying conditions requires learning a temporally parameterized family of operators rather than a single static mapping, TAM-FNO moves beyond direct temporal concatenation and introduces a spatio-temporal decoupled dual-track architecture. In this framework, the spatial SSF is processed exclusively by the FNO's 2D-FFT backbone to extract spatial wave features with limited contamination from temporally broadcast signals, while the temporal state is encoded through an independent multi-frequency Fourier pathway. These two streams are then integrated using FiLM [29], which allows the temporal variables to adaptively reweight and shift spatial feature channels. By separating spatial feature extraction from temporal modulation, TAM-FNO reduces the contamination introduced by direct concatenation while maintaining sensitivity to macroscopic environmental shifts. Numerical experiments on a South China Sea dataset demonstrate that TAM-FNO consistently improves all reported metrics compared with the static and concatenation-based variants.

II. Problem Statement

A. Problem Formulation for Time-Varying Ocean Acoustic Propagation

In underwater acoustics, the complex sound pressure field $p(\mathbf{x})$ in a range-dependent ocean waveguide is governed by the frequency-domain inhomogeneous Helmholtz equation [30]:

$$\left[\nabla^2 + k_t^2(\mathbf{x})\right] p_t(\mathbf{x}) = -S(\mathbf{x}), \tag{1}$$

where $\mathbf{x} = (r, z)$ denotes the spatial coordinates, $k_t(\mathbf{x}) = \omega / c_t(\mathbf{x})$ is the acoustic wavenumber at temporal snapshot t, and $S(\mathbf{x})$ represents the acoustic source. Here, the sound speed field $c_t(\mathbf{x})$ varies with seasonal thermodynamic conditions. Therefore, predicting the transmission loss (TL $= -20 \log_{10} |p/p_0|$) from the SSF is a task of solving the

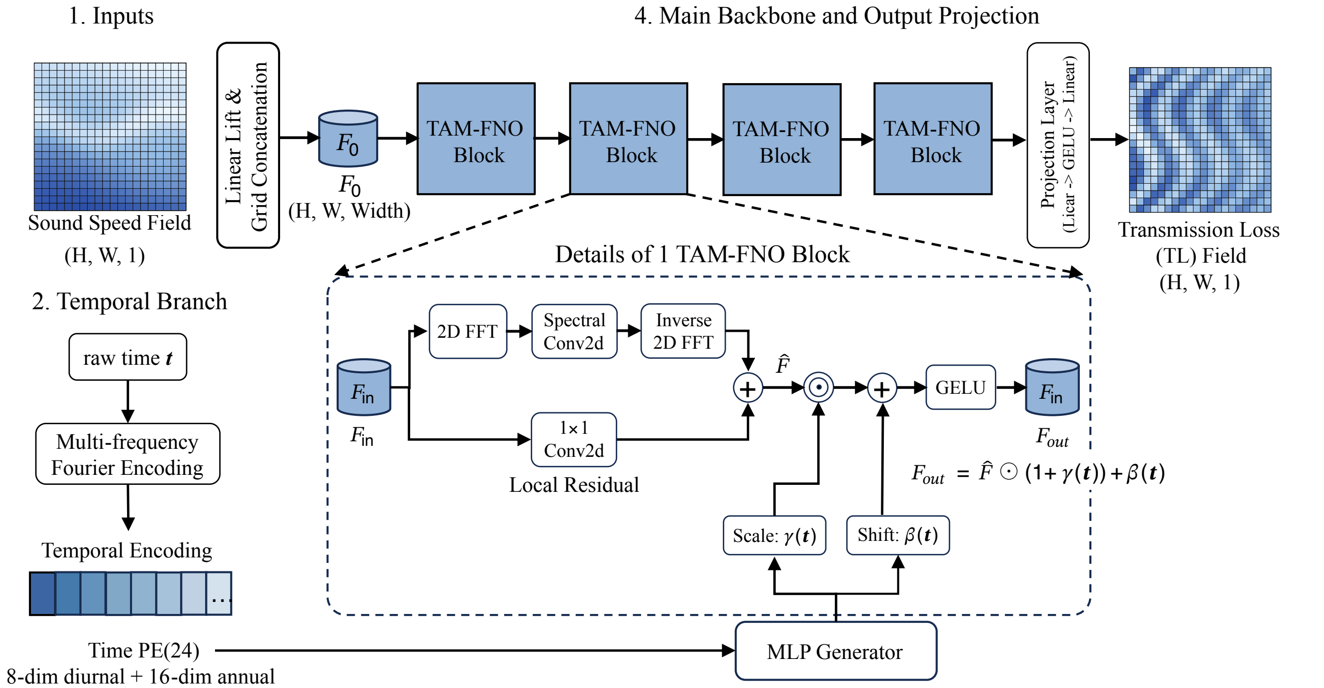

Fig. 1. The proposed TAM-FNO architecture. The spatial branch receives the SSF and maps it to a latent tensor that is processed by a four-block FNO backbone. In parallel, the temporal branch takes raw time $\mathbf{t}$, applies multi-frequency Fourier encoding, and feeds the resulting descriptor to the MLP generator. The generator produces temporal modulation terms $\gamma(\mathbf{t})$ and $\beta(\mathbf{t})$ before the final output projection to the TL prediction.

Helmholtz equation at different temporal snapshots. While traditional operator learning often assumes a static operator $\mathcal{G} : a \mapsto u$ for a fixed environment [9], [17], introducing the temporal variable t leads the model to approximate a family of operators $\mathcal{G}_t$: SSF $\rightarrow$ TL indexed by time. Thus, the task is recast as learning a time-parameterized operator family $\mathcal{G}_t$.

B. Limitation of Direct Concatenation

The vanilla FNO model [9] attempts to learn a single, static operator $\mathcal{G}$ to approximate a dynamic physical process. Adding temporal information by straightforward concatenation does not fully resolve this issue. Because t represents a global condition, broadcasting it across the 2D grid creates a spatially invariant feature map dominated by low-frequency content, which can make temporal conditioning difficult to use efficiently. At the same time, after early linear mixing, this global signal can be progressively washed out by deeper spatially varying activations, weakening temporal conditioning. These two effects motivate a decoupled architecture that separates spatial feature extraction from temporal control.

III. Proposed Method

A. TAM-FNO Architecture

As shown in Fig. 1, TAM-FNO follows a dual-track design that separates spatial representation learning from temporal conditioning. (1) The spatial branch receives the SSF as a 2D input and maps it, through the lifting stage, to a width-dimensional latent tensor F_0. (2) The temporal branch receives the raw time $\mathbf{t}$ and transforms it into a multi-frequency Fourier descriptor $\phi(\mathbf{t})$ that captures both diurnal and annual periodicity. (3) This descriptor is fed to a temporal MLP generator, which produces the modulation terms shown in Fig. 1 as $\gamma(\mathbf{t})$ and $\beta(\mathbf{t})$. (4) The main backbone propagates F_0 through four TAM-FNO blocks, where each block combines spectral convolution, local residual mixing, and temporal affine modulation, and an output projection layer maps the modulated latent field to the TL prediction. This ordering preserves a clean spatial input pathway while allowing the temporal state to steer the learned operator through block-wise modulation.

B. Multi-frequency Fourier Temporal Encoding

Rather than feeding a raw scalar directly to the network, we first map the temporal state to a periodic descriptor that can represent recurring diurnal and seasonal patterns. Let $\mathbf{t}$ denote the discrete time index associated with each SSF/TL pair, N_d the number of samples within one day, and N_y the number of samples within one year. We decompose the time index into daily and annual cycles and encode them using multi-frequency sine and cosine functions:

$$\mathbf{f}_{\text{day}}(\mathbf{t}) = \left[\sin\left(2\pi k \frac{\mathbf{t} \bmod N_d}{N_d}\right), \cos\left(2\pi k \frac{\mathbf{t} \bmod N_d}{N_d}\right)\right]_{k=1}^{K_d}, \tag{2}$$

$$\mathbf{f}_{\text{year}}(\mathbf{t}) = \left[\sin\left(2\pi k \frac{\mathbf{t}}{N_y}\right), \cos\left(2\pi k \frac{\mathbf{t}}{N_y}\right)\right]_{k=1}^{K_y}. \tag{3}$$

The encoded temporal descriptor is written as

$$\phi(\mathbf{t}) = [\mathbf{f}_{\text{day}}(\mathbf{t}), \mathbf{f}_{\text{year}}(\mathbf{t})]. \tag{4}$$

This Fourier encoding serves two purposes. First, it maps the temporal sequence into a periodic feature space, helping the model represent recurring diurnal and seasonal patterns more effectively. Second, it provides a richer temporal representation

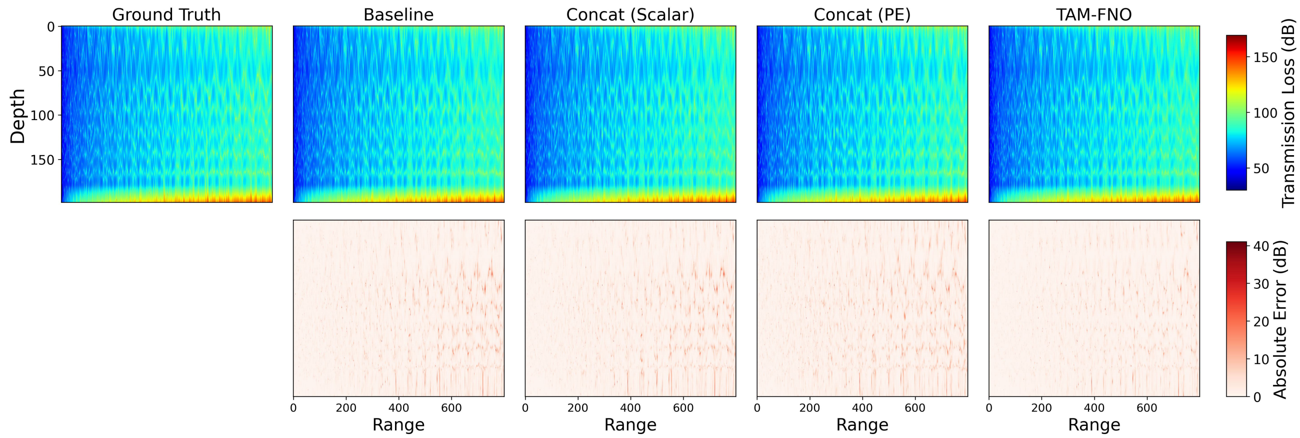

Fig. 2. Visualization of generated transmission losses (TLs). Top row: predicted TL fields; bottom row: absolute error maps. From left to right: ground truth, Baseline, Concat (Scalar), Concat (PE), and TAM-FNO.

for the MLP, especially for periodic and non-linear temporal variations [32], [33]. In our implementation, we use $N_d = 8$, $N_y = 2920$, $K_d = 4$, and $K_y = 8$, resulting in a 24-dimensional descriptor composed of 8 diurnal channels and 16 annual channels.

C. Temporally Affine Modulation and Interpretation

After each spatial layer extracts features, the temporal instruction generates a scaling coefficient $\gamma(\mathbf{t})$ and a shifting coefficient $\beta(\mathbf{t})$. Like a broadcast, these apply a global affine transformation to the spatial feature channels:

$$F_{\text{out}} = \text{GELU}(\hat{F} \odot (1 + \gamma(\mathbf{t})) + \beta(\mathbf{t})), \tag{5}$$

The residual formulation $(1+\gamma)$ ensures that at the initialization phase, the model safely degenerates into a pure 2D-FNO, preventing early training collapse.

The TAM-FNO architecture separates spatial feature extraction from temporal conditioning through its design:

1) $\hat{F}$ as Spatial Features: The intermediate feature map $\hat{F}$ extracted by the FNO backbone primarily encodes spatial information from the SSF input. Since the temporal pathway is handled separately, the backbone can focus on modeling spatial structures before temporal conditioning is injected.

2) $\gamma(\mathbf{t})$ as Channel-wise Reweighting: The multiplicative term $\gamma(\mathbf{t})$ provides channel-wise scaling conditioned on the temporal state. In this way, the temporal pathway can reweight the responses of different spatial feature channels under different environmental conditions, allowing the model to adapt its representation across seasonal variations without altering the spatial input format.

3) $\beta(\mathbf{t})$ as Channel-wise Offset: The additive term $\beta(\mathbf{t})$ introduces channel-wise offsets in the latent space. This additive modulation helps adjust the baseline of intermediate feature responses under changing ocean conditions and complements the multiplicative effect of $\gamma(\mathbf{t})$.

D. Training and Inference

During training, each SSF-time pair is processed by the spatial and temporal branches simultaneously. The spatial branch extracts latent spatial features from the SSF, while the temporal branch maps the same time index $\mathbf{t}$ to the encoded descriptor $\phi(\mathbf{t})$ and then to affine modulation terms for each TAM-FNO block. The network is optimized end-to-end against the reference TL using an H^1-style objective that combines field mismatch with first-order finite-difference mismatch [17]. SSF and TL are normalized using training-set statistics only.

During inference, a new SSF and its associated time index are passed through the same two-branch pipeline. The temporal branch generates the modulation terms, the backbone applies them block by block, and the output head produces the TL prediction in a single forward pass, avoiding repeated RAM simulations for different temporal conditions.

IV. Experiments & Results

A. Experimental Settings

We use a real-world South China Sea sound speed field dataset containing 2920 samples spanning a full year, sampled 8 times per day to capture both diurnal and seasonal variability [34]. Each SSF/TL pair is discretized on a 199×800 range-depth grid, and the TL labels are generated by the standard Range-dependent Acoustic Model (RAM) solver [3]. We randomly split the 2920 samples into 2336 training samples and 584 test samples using a fixed random seed of 2025.

All models use the same FNO backbone with $T = 4$ Fourier layers. Following the notation of Li et al. [9], the Fourier integral operator retains $k_{\max,z} = 32$ and $k_{\max,r} = 128$ modes in the depth and range directions, respectively, and the lifted feature dimension is set to $d_v = 64$; a 20-grid-point zero padding is applied before the Fourier layers. For TAM-FNO, raw time is mapped to the 24-dimensional descriptor $\phi(\mathbf{t})$, and each backbone block is conditioned by an independent two-layer MLP head that outputs 2×64 FiLM coefficients. We train all models with Adam (learning rate 10^{-3}, weight decay 10^{-4}), a StepLR scheduler with step size 50 and decay factor 0.5, batch size 20, and 100 epochs. We compare TAM-FNO against three baselines: a static 2D-FNO without temporal input (Baseline), a scalar-time concatenation variant that appends the raw temporal variable to the input channels (Concat (Scalar)), and a Fourier-encoding concatenation variant that appends the same 24-dimensional temporal descriptor to the input channels (Concat (PE)). Performance is evaluated using

TABLE I
OVERALL PERFORMANCE COMPARISON ON THE FULL TEST SET

Model	**RMSE (dB)**	**Rel. L2**	**SSIM (↑)**	**PCC (↑)**
Baseline	2.3296	0.0290	0.9220	0.9839
Concat (Scalar)	2.3249	0.0290	0.9222	0.9839
Concat (PE)	2.4847	0.0310	0.9132	0.9820
TAM-FNO (Ours)	**1.9642**	**0.0245**	**0.9446**	**0.9881**

Root Mean Square Error (RMSE), Relative L2 Error (Rel. L2), Structural Similarity Index Measure (SSIM), and Pearson Correlation Coefficient (PCC), where RMSE and Rel. L2 quantify numerical error, while SSIM and PCC reflect structural fidelity and correlation between predicted and reference TL fields [17], [42].

B. Performance Comparison

Table I and Fig. 2 summarize the quantitative and visual comparisons. TAM-FNO achieves the best values on all four metrics. Compared with the strongest non-FiLM baseline, it reduces RMSE by about 15.5%, lowers Rel. L2, and improves SSIM and PCC. The selected visual example also shows lower absolute-error intensity for TAM-FNO.

Beyond the global metrics, Fig. 2 provides a case-level view of the spatial error distribution. The three non-FiLM variants recover the large-scale TL structure, but their residual maps contain more visible streak-like errors in regions with stronger range-depth variation. TAM-FNO produces a cleaner absolute-error map, with weaker residual concentration and fewer high-error structures in the selected test case. This suggests that temporal affine modulation improves not only the averaged scores but also the local consistency of the predicted TL field.

Fig. 3 further shows the daily RMSE trend across the full year. TAM-FNO remains lower than the compared models across the year.

C. Visualizing Decoupling & Perturbation via Feature Interception

To examine how the latent space responds to temporal conditioning, we design a feature interception visualization experiment. We isolate a single SSF from the test set and keep it constant as the spatial input across all inference passes. We then inject two representative extreme time conditions $\mathbf{t}$, namely Winter and Summer, which exhibit the clearest seasonal contrast. By halting the forward pass immediately after the first Spectral Convolution layer (Block 0), we extract the intermediate activation tensors. This allows us to separate

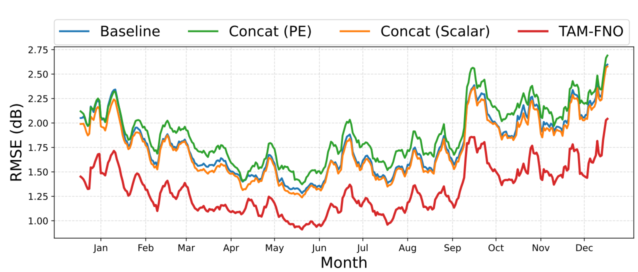

Fig. 3. Daily RMSE tracking across the full year. TAM-FNO remains lower than the compared models across the year.

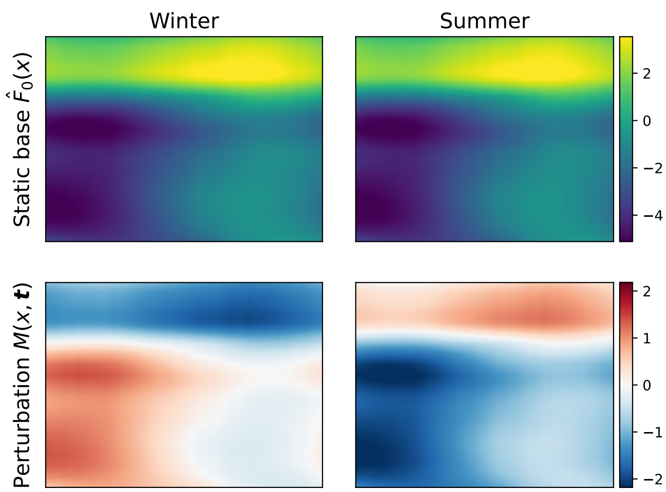

Fig. 4. Block 0 feature interception under Winter and Summer conditions. Top row: static spatial bases $\hat{F}_0(x)$. Bottom row: temporal perturbation field $M(x, \mathbf{t})$.

the output into a spatial feature map $\hat{F}_0(x)$ and a temporally conditioned perturbation field $M(x, \mathbf{t}) = \gamma(\mathbf{t}) \odot \hat{F}_0(x) + \beta(\mathbf{t})$ before deeper non-linear mixing.

The results in Fig. 4 support the intended decoupling behavior:

1) Successful Decoupling of Spatial Feature: The feature maps $\hat{F}_0(x)$ remain visually consistent across the Winter and Summer conditions, indicating decoupling between spatial feature extraction and temporal conditioning at this stage.

2) Dynamic Perturbation Field: By independently visualizing $M(x, \mathbf{t})$, we observe that the temporal modulation changes substantially across seasons. The Winter and Summer perturbation fields exhibit nearly opposite spatial patterns: positive responses in Winter are mainly concentrated in the lower-left region, while Summer shows positive responses in the upper region and stronger negative responses over the lower-left region. This indicates that the temporal pathway can adjust intermediate feature responses according to seasonal conditions while preserving the same spatial input.

V. CONCLUSION

TAM-FNO provides a practical architecture for introducing temporal awareness into FNO. By employing a spatio-temporal dual-track design and affine modulation, it improves how temporal information is incorporated compared with simple concatenation. In the reported experiments, TAM-FNO consistently outperforms the static and concatenation-based variants on all reported metrics, although the performance gap is moderate rather than overwhelming. The perturbation interception visualization further indicates that the network can preserve spatial feature extraction while adjusting intermediate responses according to time, supporting the effectiveness of temporally conditioned operator learning for dynamic underwater acoustic prediction.

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